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(54) HOUSING, SEMICONDUCTOR MODULE COMPRISING A HOUSING, AND METHOD FOR ASSEMBLING A SEMICONDUCTOR MODULE

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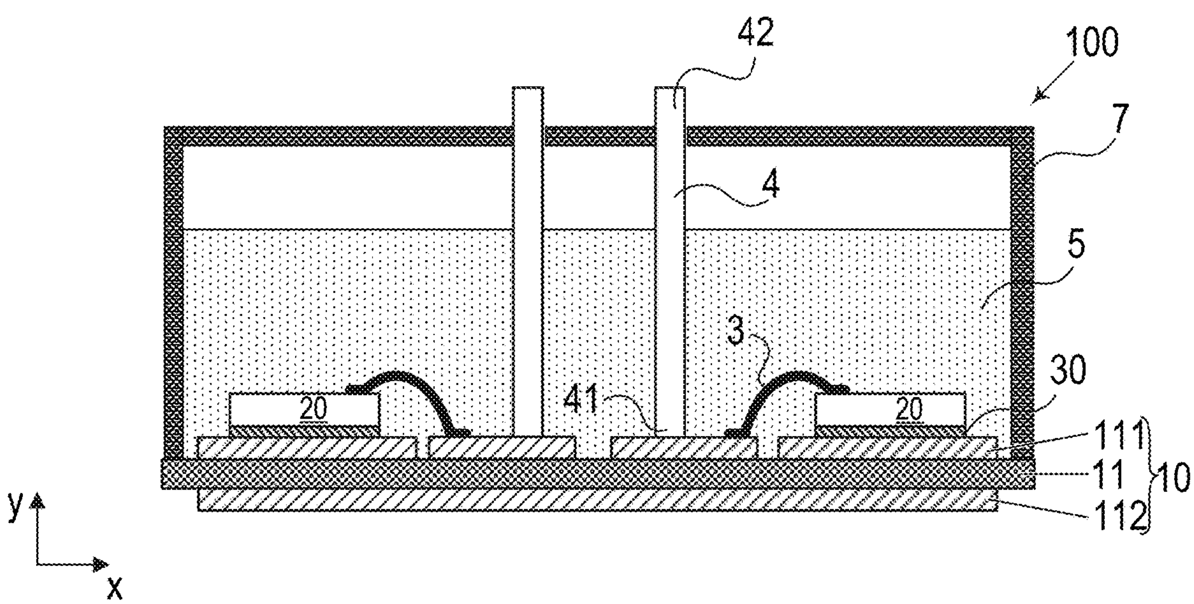
(57) **ABSTRACT**

A housing for a semiconductor module comprises a first plurality n of side elements, and a second plurality m of hinge portions, wherein n=m+1, in an unassembled state of the housing, the side elements are arranged successively in one row and each of the hinge portions is arranged to connect two directly neighboring side elements, each of the side elements is foldable with respect to at least one other side element about a respective folding axis, and the side elements are configured to form a closed frame in an assembled state of the housing.



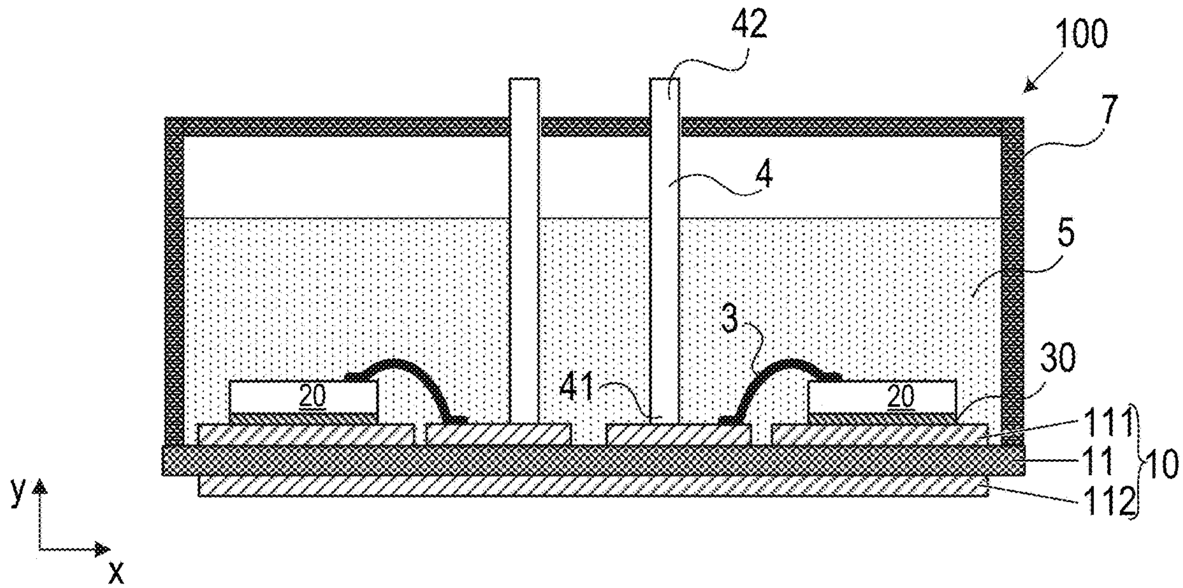

42
100
7
4
5
3
30
41
20
20
111
11
10
112
y
x

FIG 1

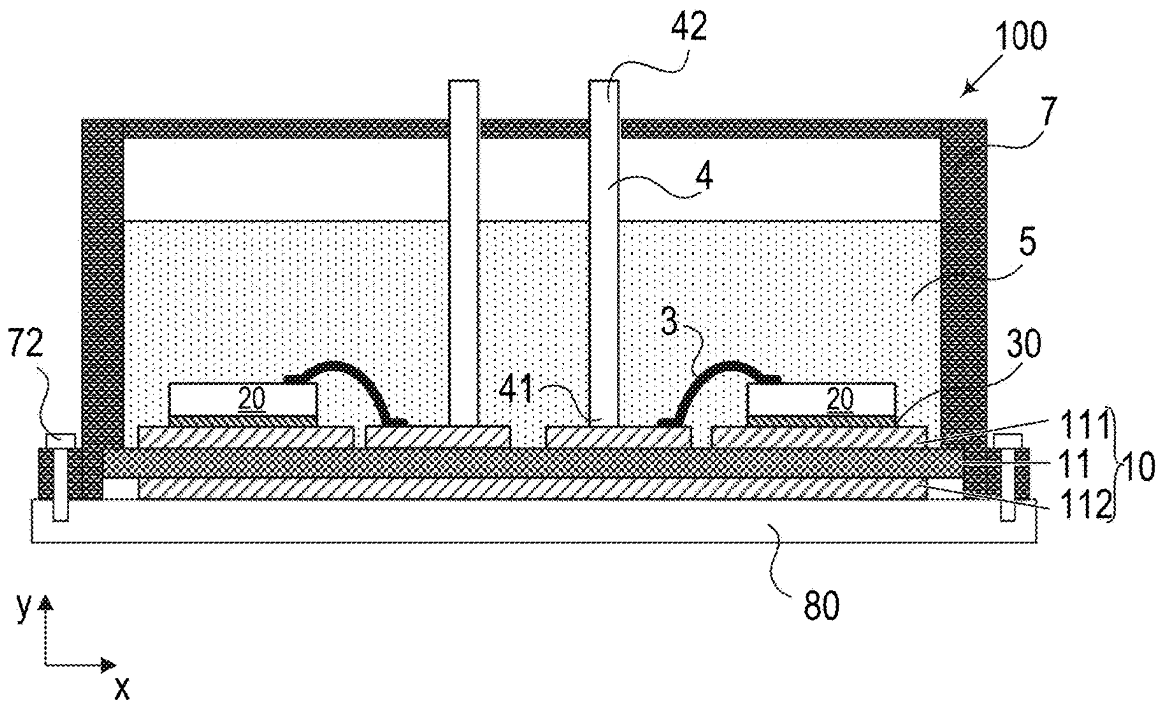

42
100
7
4
5
72
3
30
41
20
20
111
11
10
112
80
y
x

FIG 2

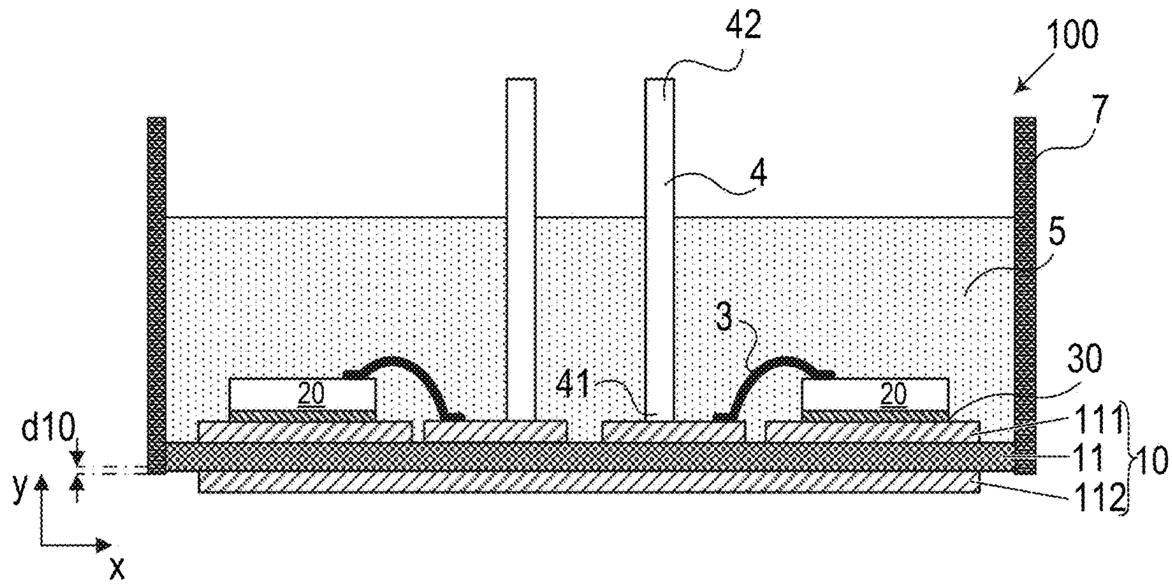


FIG 3

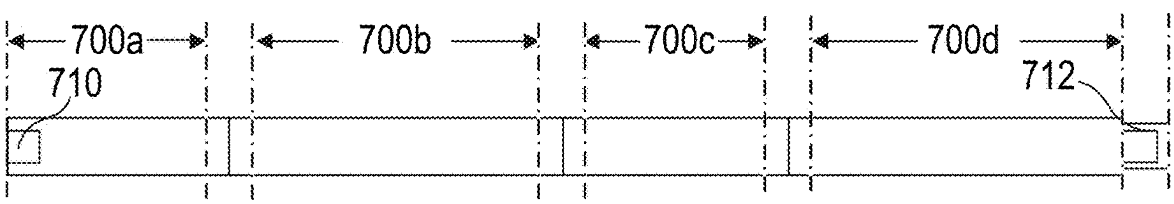


FIG 4A

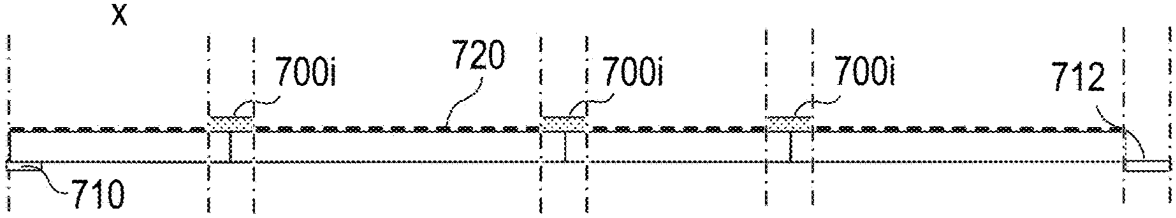


FIG 4B

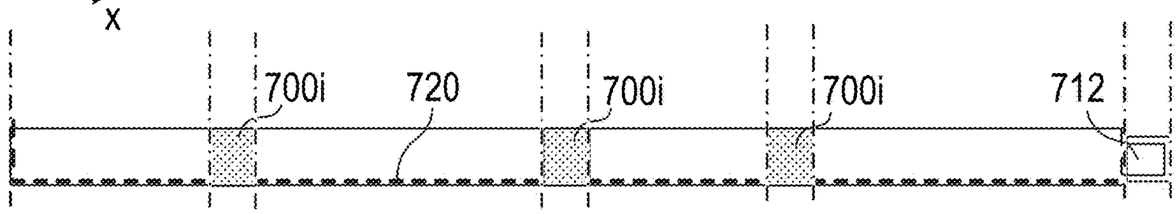


FIG 4C

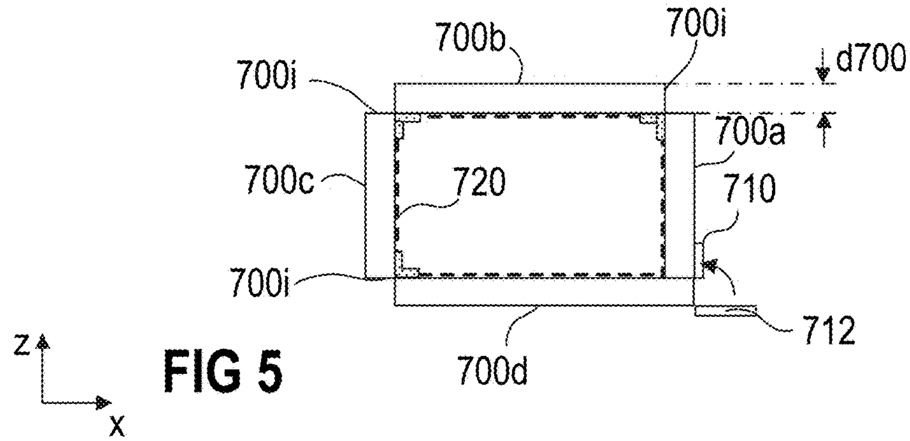


FIG 5

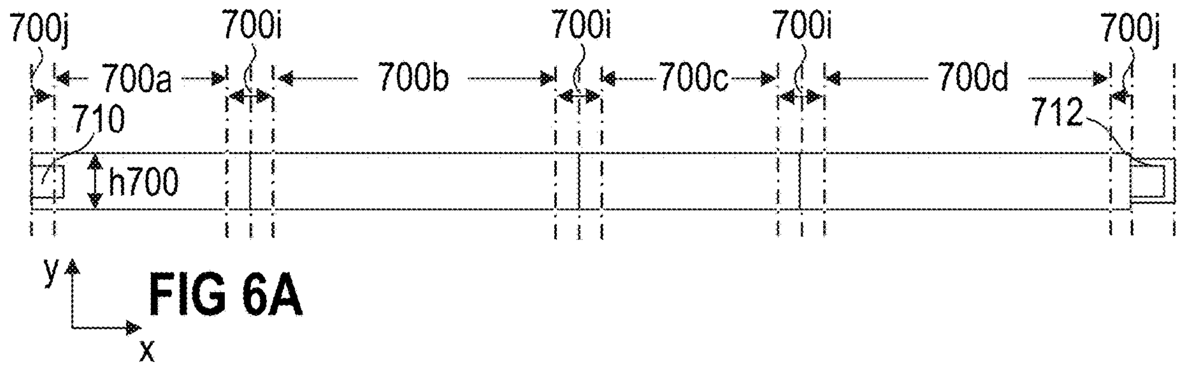


FIG 6A

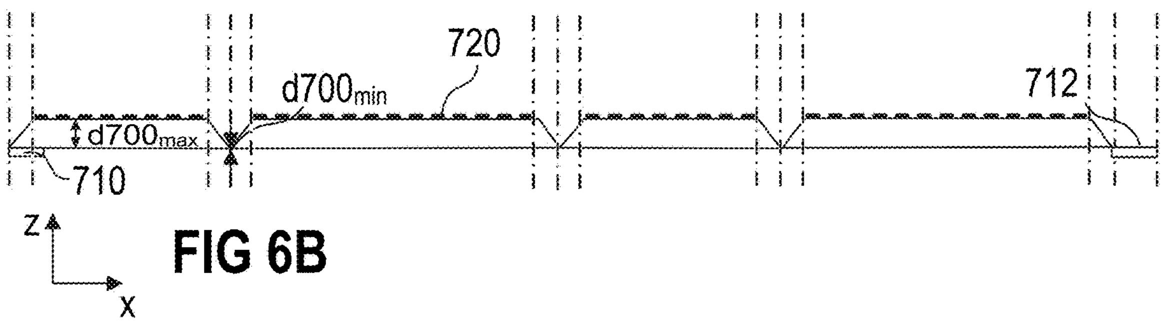


FIG 6B

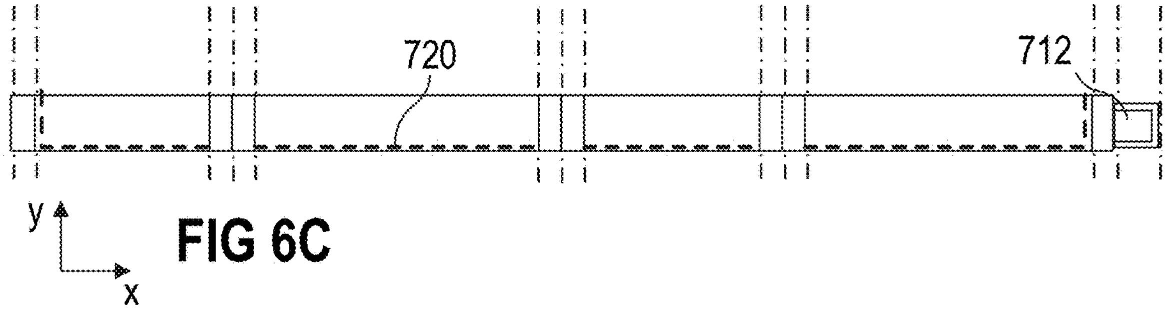


FIG 6C

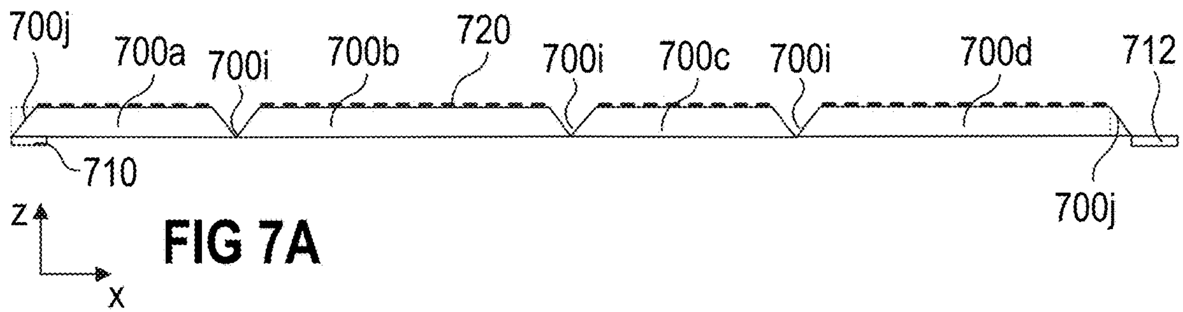


FIG 7A

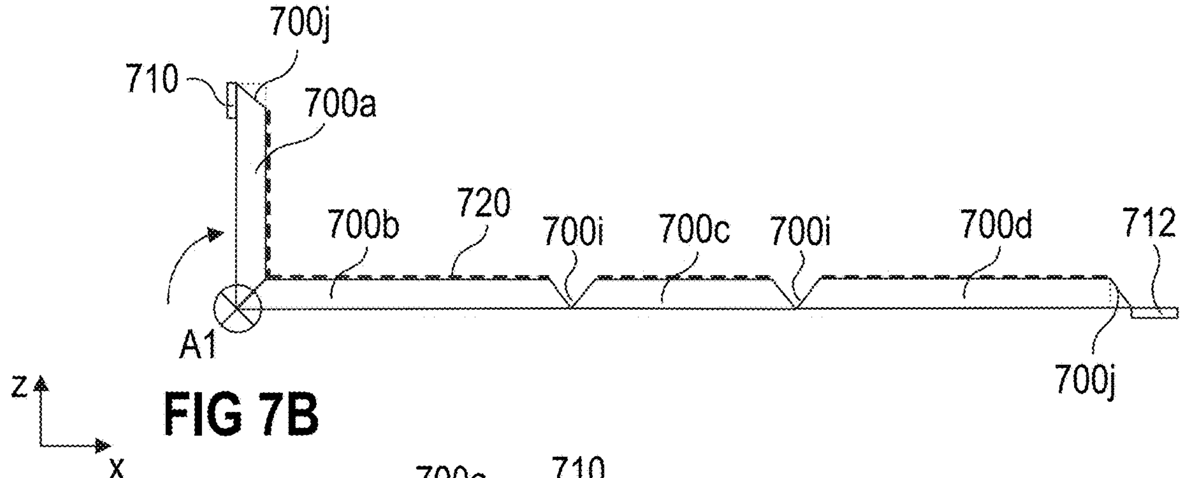


FIG 7B

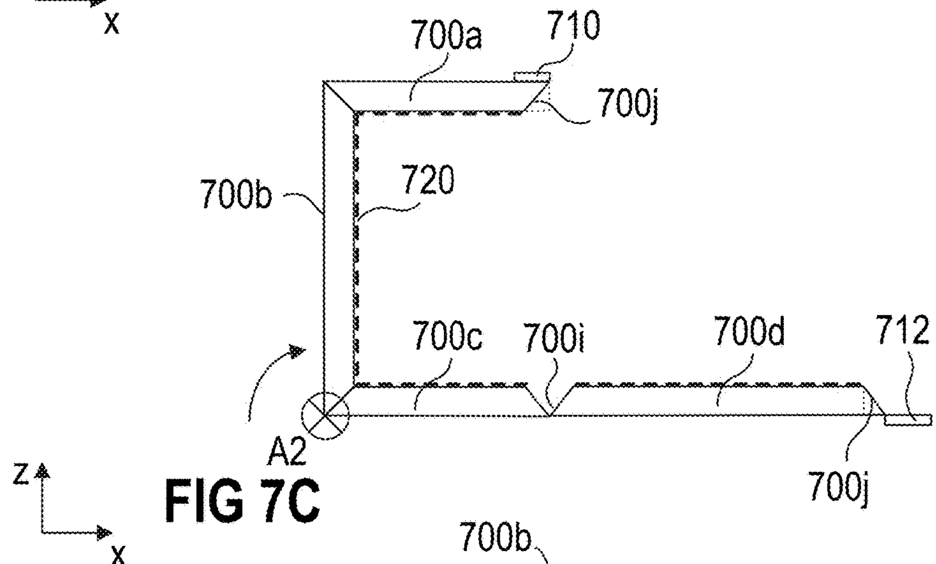


FIG 7C

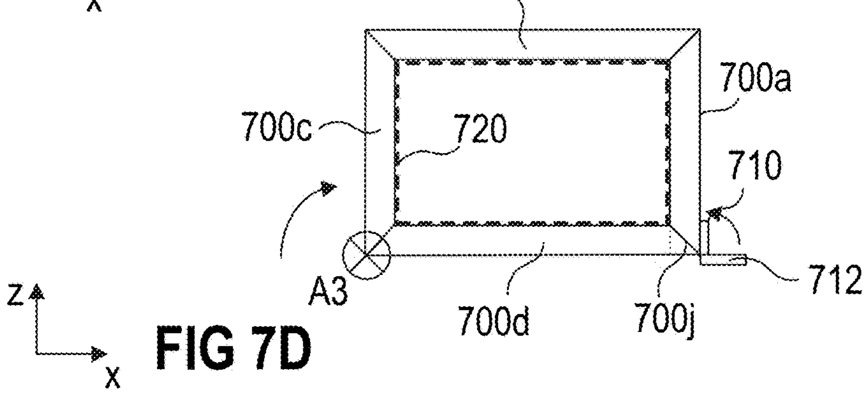


FIG 7D

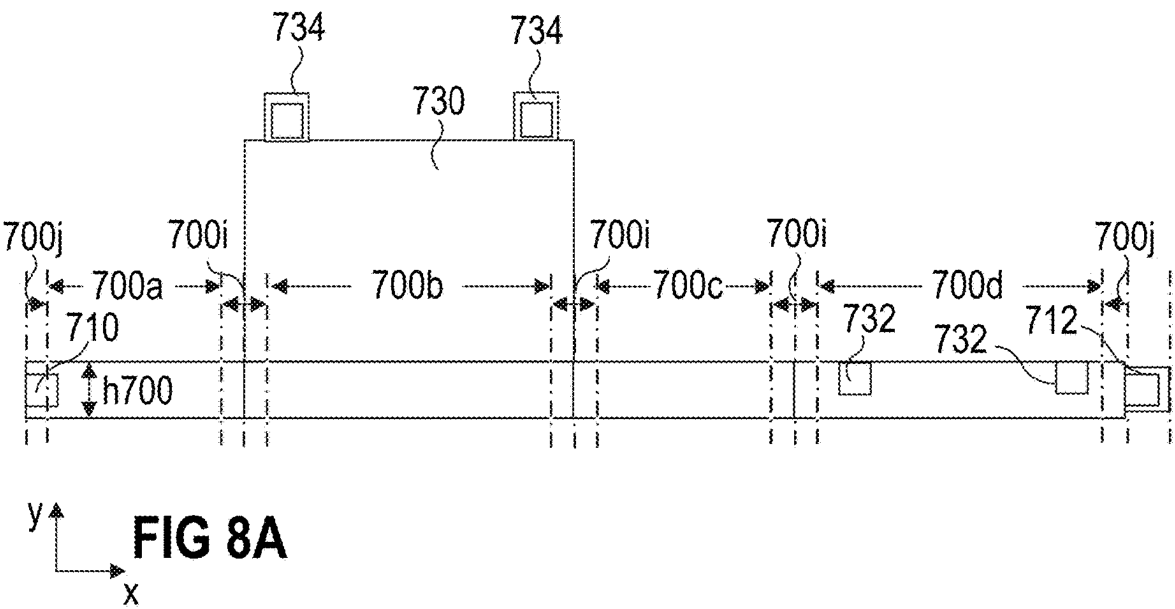


FIG 8A

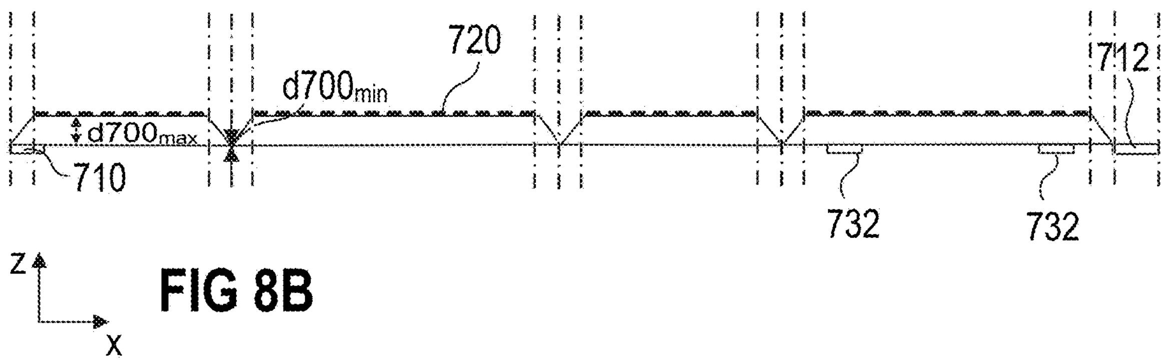


FIG 8B

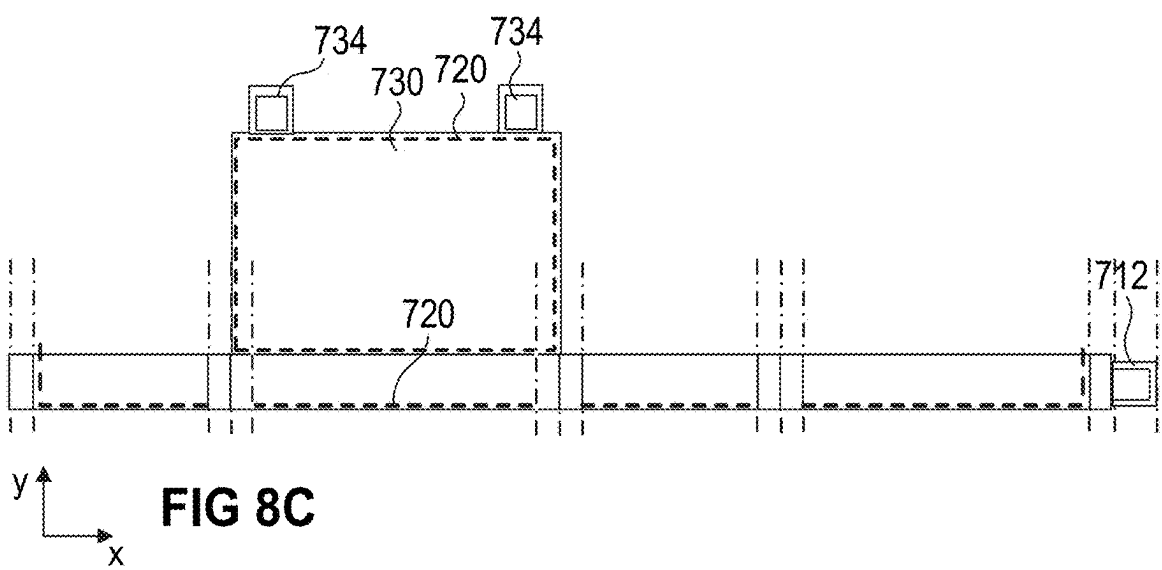


FIG 8C

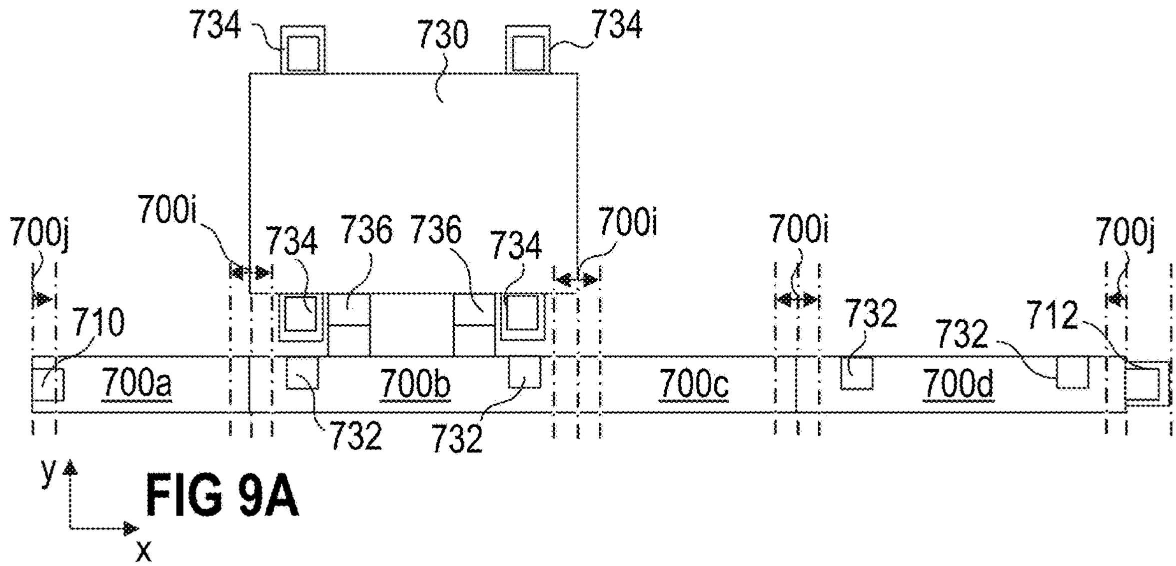


FIG 9A

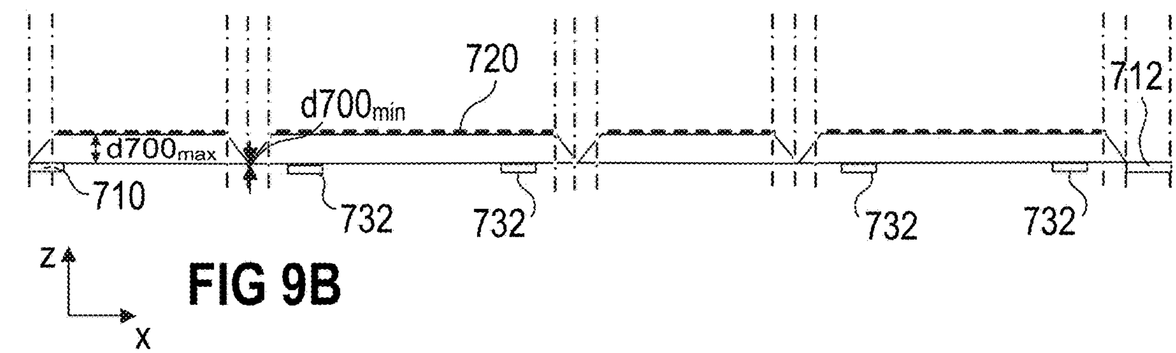


FIG 9B

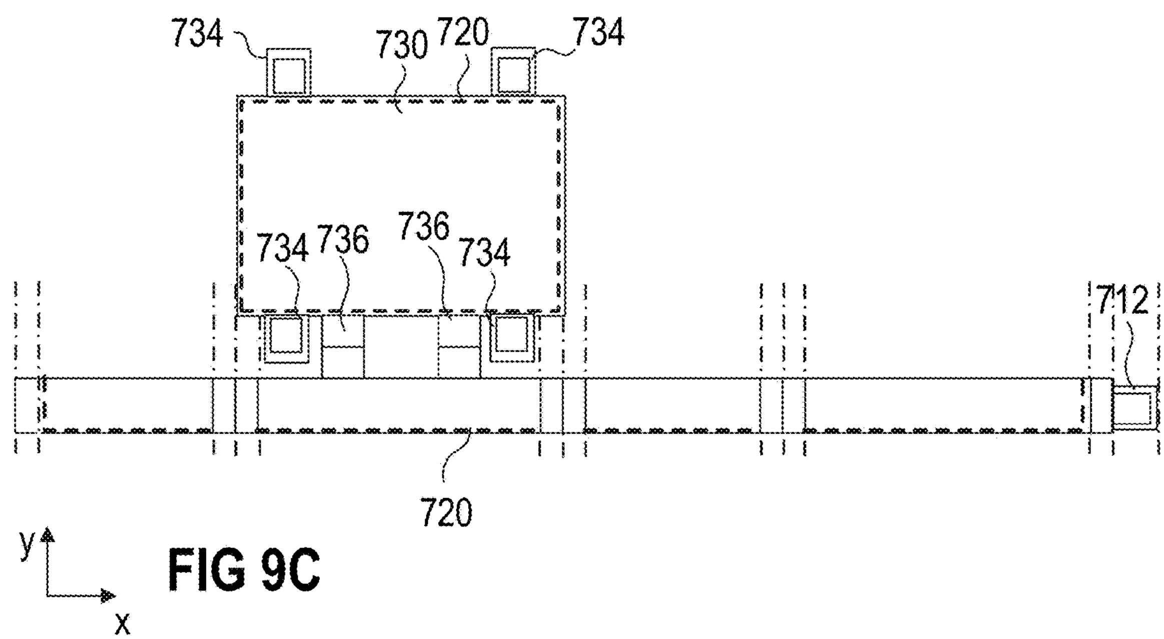


FIG 9C

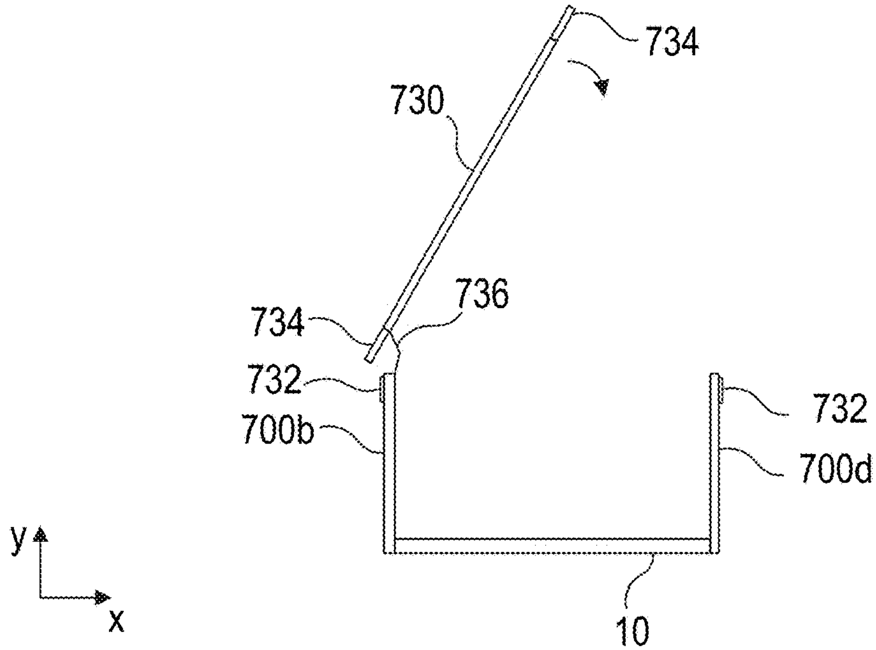


FIG 10A

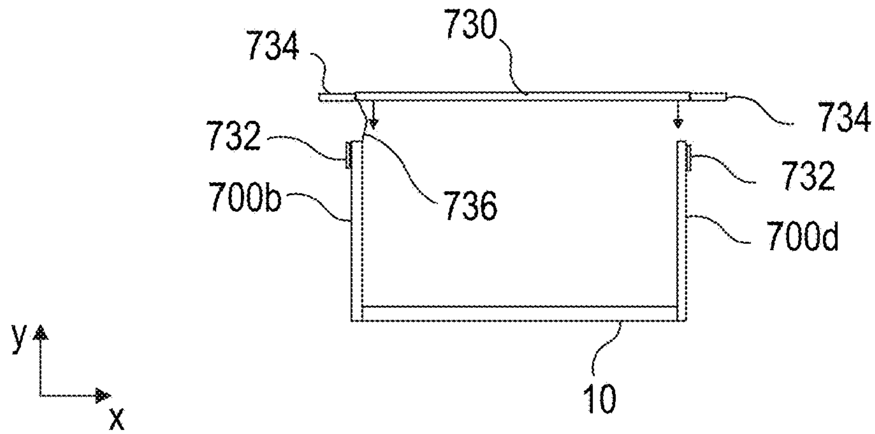


FIG 10B

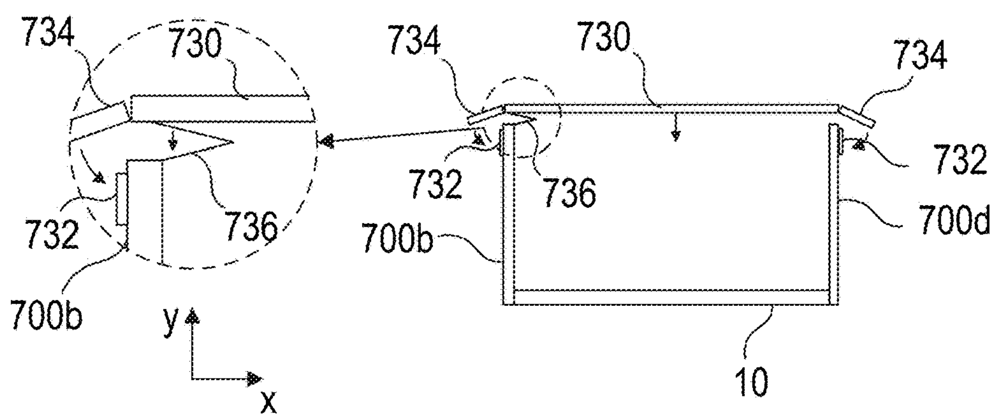


FIG 10C

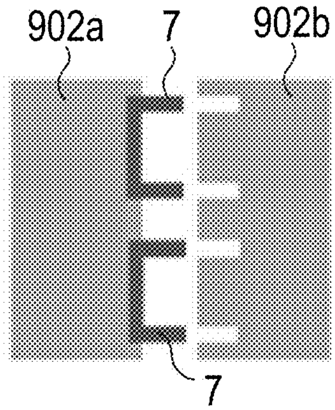


FIG 11A

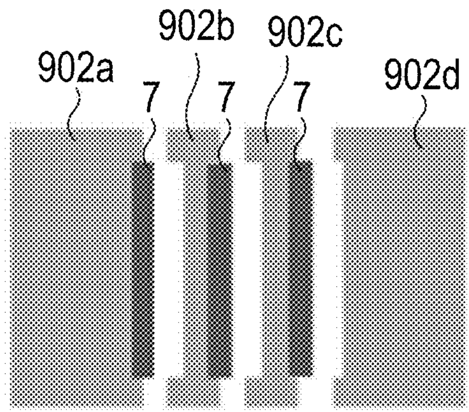


FIG 11B

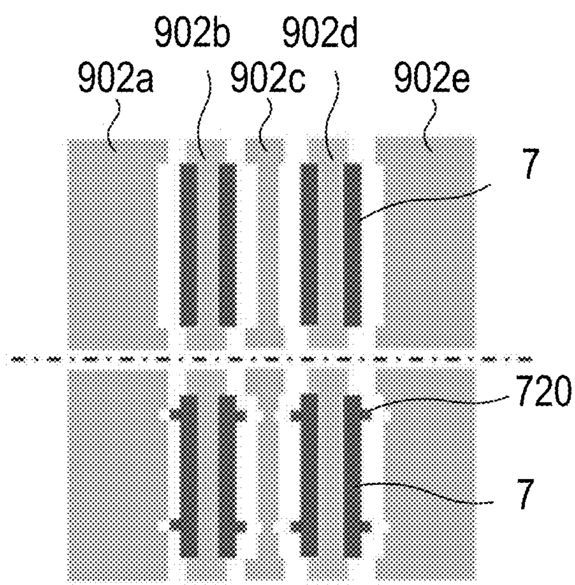


FIG 12

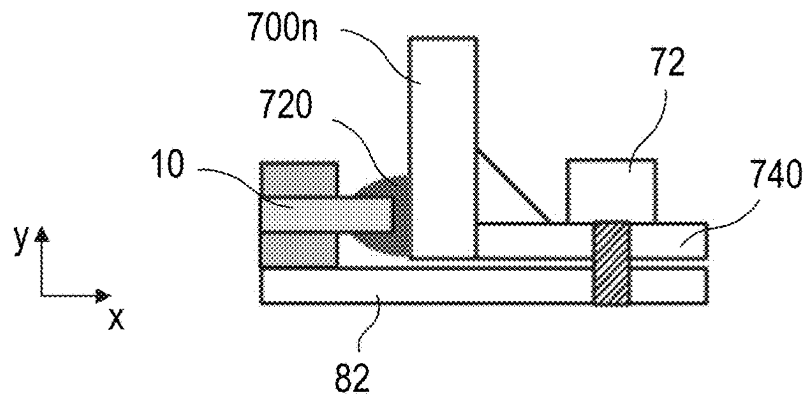


FIG 13

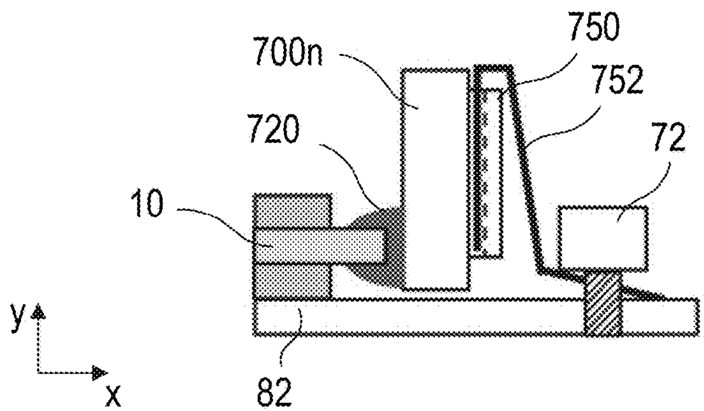


FIG 14

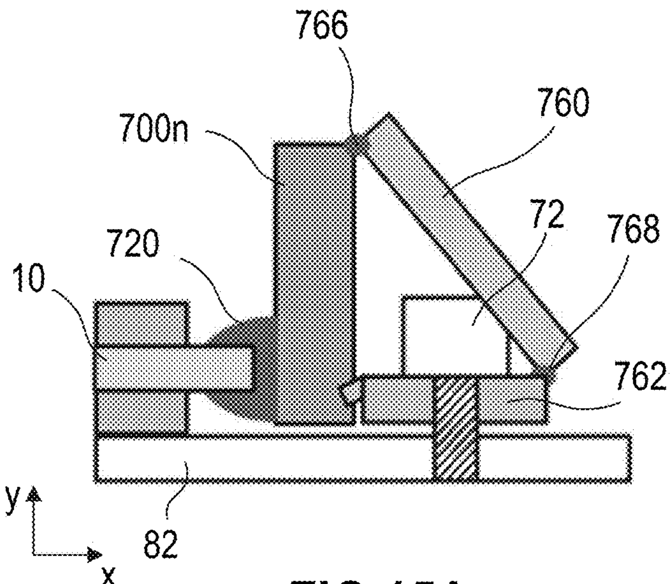


FIG 15A

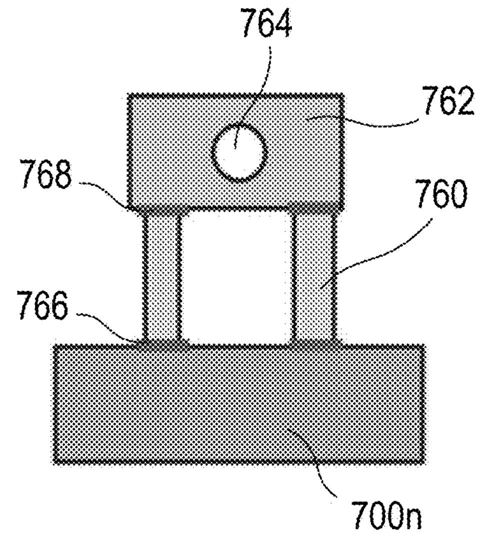


FIG 15B

HOUSING, SEMICONDUCTOR MODULE COMPRISING A HOUSING, AND METHOD FOR ASSEMBLING A SEMICONDUCTOR MODULE

TECHNICAL FIELD

[0001] The instant disclosure relates to a housing, a semiconductor module comprising a housing, and a method for assembling a semiconductor module.

BACKGROUND

[0002] Power semiconductor module arrangements often include at least one substrate arranged in a housing. A semiconductor arrangement including a plurality of controllable semiconductor elements (e.g., two IGBTs in a half-bridge configuration) or non-controllable semiconductor elements (e.g., arrangements of diodes) is arranged on each of the at least one substrate. Each substrate usually comprises a substrate layer (e.g., a ceramic layer), a first metallization layer deposited on a first side of the substrate layer and a second metallization layer deposited on a second side of the substrate layer. The controllable semiconductor elements are mounted, for example, on the first metallization layer. The second metallization layer may optionally be attached to a base plate.

[0003] The housing is usually glued to the substrate or to a base plate. One of the functions of the housing and the glue is to seal the volume inside the housing and to contain a dielectric gel (e.g., encapsulant) inside this volume. In order to sufficiently seal the volume inside of the housing, for many applications additional mechanical connection elements may be used to form a connection between the housing and the substrate or base plate in order to hold the assembly together before curing of the sealing (glue). This usually requires the presence of a base plate such that the housing can be attached to the base plate, e.g., by means of screws or rivets, thereby pressing the housing on the substrate and/or base plate. The sealing of housings in base plate less semiconductor modules is often not satisfactory.

[0004] There is a need for a housing that allows to form a satisfactory seal between the housing and a substrate even in base plate less semiconductor module arrangements, as well as for a semiconductor module comprising such a housing, and a method for assembling a semiconductor module.

SUMMARY

[0005] A housing for a semiconductor module includes a first plurality n of side elements, and a second plurality m of hinge portions, wherein n=m+1, in an unassembled state of the housing, the side elements are arranged successively in one row and each of the hinge portions is arranged to connect two directly neighboring side elements, each of the side elements is foldable with respect to at least one other side element about a respective folding axis, and the side elements are configured to form a closed frame in an assembled state of the housing.

[0006] A semiconductor module includes a housing in its assembled shape, and a substrate, wherein the housing forms a closed frame around the substrate and exerts pressure on each of a plurality of side surfaces of the substrate along the circumference of the substrate.

[0007] A method for assembling a semiconductor module includes folding a housing around a substrate by bending the plurality of side elements about respective bending axes until the side elements form a closed frame around the substrate and exert pressure on each of a plurality of side surfaces of the substrate along the circumference of the substrate.

[0008] The invention may be better understood with reference to the following drawings and the description. The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the invention. Moreover, in the figures, like referenced numerals designate corresponding parts throughout the different views.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a cross-sectional view of a semiconductor module without a base plate.

[0010] FIG. **2** is a cross-sectional view of a semiconductor module comprising a base plate.

[0011] FIG. **3** is a cross-sectional view of a semiconductor module comprising a housing according to embodiments of the disclosure.

[0012] FIG. **4**, including FIGS. **4**A-**4**C, schematically illustrates different views of a housing according to embodiments of the disclosure.

[0013] FIG. **5** schematically illustrates a top view of the housing of FIG. **4** in an assembled state.

[0014] FIG. **6**, including FIGS. **6**A-**6**C, schematically illustrates different views of another housing according to embodiments of the disclosure.

[0015] FIG. **7**, including FIGS. **7**A-**7**D, schematically illustrates steps of a method for assembling a semiconductor module comprising a housing according to embodiments of the disclosure.

[0016] FIG. **8**, including FIGS. **8**A-**8**C, schematically illustrates different views of a housing according to further embodiments of the disclosure.

[0017] FIG. **9**, including FIGS. **9**A-**9**C, schematically illustrates different views of a housing according to even further embodiments of the disclosure.

[0018] FIG. **10**, including FIGS. **10**A-**10**C, schematically illustrates a cross-sectional view of the housing of FIG. **7** during assembly.

[0019] FIG. **11**, including FIGS. **11**A and **11**B, schematically illustrates a conventional molding form and an exemplary molding form for forming a housing according to embodiments of the disclosure.

[0020] FIG. **12** schematically illustrates steps of a method for forming a housing according to embodiments of the disclosure.

[0021] FIG. **13** schematically illustrates a cross-sectional view of a section of a housing according to even further embodiments of the disclosure.

[0022] FIG. **14** schematically illustrates a cross-sectional view of a section of a housing according to even further embodiments of the disclosure.

[0023] FIG. **15**, including FIGS. **15**A and **15**B, schematically illustrates a cross-sectional view of a section of a housing and a top view of a section of a housing according to even further embodiments of the disclosure.

DETAILED DESCRIPTION

[0024] In the following detailed description, reference is made to the accompanying drawings. The drawings show specific examples in which the invention may be practiced. It is to be understood that the features and principles described with respect to the various examples may be combined with each other, unless specifically noted otherwise. In the description, as well as in the claims, designations of certain elements as “first element”, “second element”, “third element” etc. are not to be understood as enumerative. Instead, such designations serve solely to address different “elements”. That is, e.g., the existence of a “third element” does not require the existence of a “first element” and a “second element”. An electrical line or electrical connection as described herein may be a single electrically conductive element, or include at least two individual electrically conductive elements connected in series and/or parallel. Electrical lines and electrical connections may include metal and/or semiconductor material, and may be permanently electrically conductive (i.e., non-switchable). A semiconductor body as described herein may be made from (doped) semiconductor material and may be a semiconductor chip or be included in a semiconductor chip. A semiconductor body has electrically connecting pads and includes at least one semiconductor element with electrodes.

[0025] Referring to FIG. **1**, a cross-sectional view of a semiconductor module **100** is illustrated. The semiconductor module **100** includes a housing **7** and a substrate **10**. The substrate **10** includes a dielectric insulation layer **11**, a (structured) first metallization layer **111** attached to the dielectric insulation layer **11**, and a (structured) second metallization layer **112** attached to the dielectric insulation layer **11**. The dielectric insulation layer **11** is disposed between the first and second metallization layers **111**, **112**.

[0026] Each of the first and second metallization layers **111**, **112** may consist of or include one of the following materials: copper; a copper alloy; aluminum; an aluminum alloy; any other metal or alloy that remains solid during the operation of the power semiconductor module arrangement. The substrate **10** may be a ceramic substrate, that is, a substrate in which the dielectric insulation layer **11** is a ceramic, e.g., a thin ceramic layer. The ceramic may consist of or include one of the following materials: aluminum oxide; aluminum nitride; zirconium oxide; silicon nitride; boron nitride; or any other dielectric ceramic. For example, the dielectric insulation layer **11** may consist of or include one of the following materials: Al_2O_3, AlN, SiC, BeO or Si_3N_4. For instance, the substrate **10** may, e.g., be a Direct Copper Bonding (DCB) substrate, a Direct Aluminum Bonding (DAB) substrate, or an Active Metal Brazing (AMB) substrate. Further, the substrate **10** may be an Insulated Metal Substrate (IMS). An Insulated Metal Substrate generally comprises a dielectric insulation layer **11** comprising (filled) materials such as epoxy resin or polyimide, for example. The material of the dielectric insulation layer **11** may be filled with ceramic particles, for example. Such particles may comprise, e.g., SiO_2, Al_2O_3, AlN, or BN and may have a diameter of between about 1 μm and about 50 μm. The substrate **10** may also be a conventional printed circuit board (PCB) having a non-ceramic dielectric insulation layer **11**. For instance, a non-ceramic dielectric insulation layer **11** may consist of or include a cured resin.

[0027] The substrate **10** is arranged in a housing **7**. In the example illustrated in FIG. **1**, the substrate **10** forms a ground surface of the housing **7**, while the housing **7** itself solely comprises sidewalls and a cover. This is, however, only an example. It is also possible that the housing **7** further comprises a ground surface and the substrate **10** be arranged inside the housing **7**. According to another example, the substrate **10** may be mounted on a base plate. This is schematically illustrated in FIG. **2**. In some semiconductor modules **100**, more than one substrate **10** is arranged on a single base plate. The base plate may form a ground surface of the housing **7**, for example. The top of the housing **7** can either be a separate cover or lid that can be removed from the sidewalls, or may be formed integrally with the sidewalls of the housing **7**. In the latter case, the top and the sidewalls of the housing **7** may be formed as a single piece such that the top cannot be removed from the sidewalls without damaging or destroying the housing **7**.

[0028] One or more semiconductor bodies **20** may be arranged on the substrate **10**. Each of the semiconductor bodies **20** arranged on the substrate **10** may include a diode, an IGBT (Insulated-Gate Bipolar Transistor), a MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor), a JFET (Junction Field-Effect Transistor), a HEMT (High-Electron-Mobility Transistor), or any other suitable controllable or non-controllable semiconductor element.

[0029] The one or more semiconductor bodies **20** may form a semiconductor arrangement on the substrate **10**. In FIG. **1**, only two semiconductor bodies **20** are exemplarily illustrated. The second metallization layer **112** of the substrate **10** in FIG. **1** is a continuous layer. The first metallization layer **111** is a structured layer in the example illustrated in FIG. **1**. “Structured layer” means that the first metallization layer **111** is not a continuous layer, but includes recesses between different sections of the layer. Such recesses are schematically illustrated in FIG. **1**. The first metallization layer **111** in this example includes four different sections. Different semiconductor bodies **20** may be mounted to the same or to different sections of the first metallization layer **111**. Different sections of the first metallization layer may have no electrical connection or may be electrically connected to one or more other sections using, e.g., bonding wires **3**. Electrical connections **3** may also include connection plates or conductor rails, for example, to name just a few examples. The one or more semiconductor bodies **20** may be electrically and mechanically connected to the substrate **10** by an electrically conductive connection layer **30**. Such an electrically conductive connection layer may be a solder layer, a layer of an electrically conductive adhesive, or a layer of a sintered metal powder, e.g., a sintered silver powder, for example.

[0030] The semiconductor module **100** illustrated in FIG. **1** further includes terminal elements **4**. The terminal elements **4** are electrically connected to the first metallization layer **111** and provide an electrical connection between the inside and the outside of the housing **7**. The terminal elements **4** may be electrically connected to the first metallization layer **111** with a first end **41**, while a second end **42** of the terminal elements **4** protrudes out of the housing **7**. The terminal elements **4** may be electrically contacted from the outside at their second end **42**. The terminal elements **4** illustrated in FIG. **1**, however, are only examples. Terminal elements **4** may be implemented in any other way and may be arranged at any other position. For example, one or more terminal elements **4** may be arranged close to or adjacent to the sidewalls of the housing **7**. Any other suitable implementation is possible. The terminal elements **4** may consist of or include a metal such as copper, aluminum, gold, silver, or any alloys thereof, for example. The terminal elements **4**

may be electrically and mechanically connected to the substrate **10** by an electrically conductive connection layer (not specifically illustrated for the terminal elements **4**). Such an electrically conductive connection layer generally may be a solder layer, a layer of an electrically conductive adhesive, or a layer of a sintered metal powder, e.g., a sintered silver powder, for example.

[0031] Conventional semiconductor modules **100** generally further include a casting compound **5**. The casting compound **5** may consist of or include a silicone gel or may be a rigid molding compound, for example. The casting compound **5** may at least partly fill the interior of the housing **7**, thereby covering the components and electrical connections that are arranged on the substrate **10**. In the illustrated embodiment, terminal elements **4** may be partly embedded in the casting compound **5**. At least their second ends **42**, however, are not covered by the casting compound **5** and protrude from the casting compound **5** through the housing **7**, to the outside of the housing **7**. The casting compound **5** is configured to protect the components and electrical connections inside the semiconductor module **100**, in particular inside the housing **7**, from certain environmental conditions and mechanical damage.

[0032] The casting compound **5** is usually formed by filling a liquid or semi-liquid material into the housing **7**. The initially liquid or semi-liquid material is then hardened (cross-linked), e.g., by heating the semiconductor module **100**. The hardening or cross-linking process generally requires a certain amount of time. That is, the material remains liquid or semi-liquid for a certain amount of time and there is a risk that material leaks out of the housing **7** before it has been hardened to such a degree that it may no longer leak out of the housing **7**. In arrangements similar to the one that is illustrated in FIG. **1** (base plate less semiconductor modules), the housing **7** is usually glued directly on the substrate **10**. The glue which attaches the housing **7** to the substrate **10** is not only required to attach the housing **7** to the substrate **10**, but also to sufficiently seal the housing **7** to prevent material forming the encapsulant **5** from leaking out of the housing **7**. This similarly applies for semiconductor modules **100** where the housing **7** is attached to the substrate **10** in different ways instead of gluing, as well as for semiconductor modules **100** similar to the semiconductor module **100** of FIG. **2** (semiconductor modules comprising a base plate **80**). Here, the housing **7** is usually attached to the base plate **80** by means of suitable mechanical connection elements, e.g., by means of rivets or screws **72**. When attaching the housing **7** to the base plate **80**, the housing **7**, at the same time, may be pressed on the base plate **80** and/or the substrate **10**. A sealing may be provided between the housing and the base plate **80** and/or between the housing **7** and the substrate **10** in order to prevent material forming the encapsulant **5** from leaking out of the housing **7**. The arrangements as illustrated in FIGS. **1** and **2** may be sophisticated and complex and mounting the housing **7** to the substrate **10** or base plate **80** may be cumbersome. Further, production costs of the housing **7** may be comparably high.

[0033] Now referring to FIG. **3**, a semiconductor module **100** according to embodiments of the disclosure is schematically illustrated. The semiconductor module **100** may be a base plate less module, for example. That is, the substrate **10** itself may form a base surface of the housing **7**. The housing **7**, however, instead of being glued to the substrate **10** or attached to the substrate **10** by any kind of mechanical connection means, is instead simply folded around the substrate **10**. That is, the housing **7** is in contact with the side surfaces of the substrate **10**, wherein the side surfaces are surfaces of the substrate **10** extending between a top surface of the substrate **10** and a bottom surface of the substrate **10**, opposite the top surface. The top surface of the substrate **10** is a surface on which the semiconductor bodies **20** are mounted and which faces the inside of the housing **7**, when the housing **7** is mounted to the substrate **10**. The housing **7** may apply a certain amount of pressure on the side surfaces of the substrate **10** in order to securely hold the substrate **10** in a desired position with respect to the housing **7**. This will be described further by means of FIGS. **4** to **7** in the following. However, it is also contemplated that some kind of glue or sealing may be applied between the inside of the housing **7** and the side surfaces of the substrate **10**, to more effectively seal the interior and/or to more securely fix the housing **7** to the substrate **10**.

[0034] FIGS. **4**A to **4**C illustrate a housing **7** according to embodiments of the disclosure in an unmounted (unfolded state), wherein FIG. **4**A illustrates an outside of the housing **7**, FIG. **4**B illustrates a top view of the housing **7**, and FIG. **4**C illustrates an inside of the housing **7**. The housing **7** comprises a first plurality n of side elements **700*n***, a second plurality m of hinge portions **700*i***, and a fastening element **710**, **712**. The number of hinge portions **700*i*** may be one less than the number of side elements **700*n***. That is, n=m+1. In an unassembled state of the housing **7**, the side elements **700*n*** are arranged successively in one row and each of the second plurality m of hinge portions **700*i*** is arranged to connect two directly neighboring side elements **700*n*** to each other. Each of the first plurality n of side elements **700*n*** is foldable with respect to at least one other side element **700*n*** about a respective folding axis.

[0035] For example, a first outermost side element **700*a*** in the row of side elements **700*n*** (the first outermost side element **700*a*** only has one neighboring side element **700*b***) may be foldable about a first folding axis. In particular, the first outermost side element **700*a*** may be foldable with respect to a directly preceding second side element **700*b***. The respective folding axis may be located at a position between the first outermost side element **700*a*** and the second side element **700*b*** (see, e.g., first folding axis A**1** in FIG. **7**B). The second side element **700*b***, being arranged between two other side elements **700*a***, **700*c***, may be foldable with respect to each of its neighboring side elements **700*a***, **700*c*** about respective folding axes (see, e.g., first folding axis A**1** and second folding axis A**2** in FIGS. **7**B and **7**C). This similarly applies for the third side element **700*c*** which is foldable with respect to each of its neighboring side elements **700*b***, **700*d*** about respective folding axes (see, e.g., second folding axis A**2** and third folding axis A**3** in FIGS. **7**C and **7**D). In the examples described herein, the fourth side element **700*d*** is another outermost side element at an opposite end of the row of side elements **700*n*** than the first outermost side element **700*a***. The fourth outermost side element **700*d***, similar to the first outermost side element **700*a***, is foldable only with respect to one other side element, namely the third side element **700*c*** (see, e.g., third folding axis A**3** in FIG. **7**D).

[0036] As can be seen in FIGS. **5** and **7**D, for example, the first plurality n of side elements **700*n*** is configured to form a closed frame in an assembled state. The fastening element **710**, **712** is configured to connect the two outermost side elements **700**a, **700**d with each other such that the housing **7** remains in its assembled/folded shape and exerts pressure on a substrate **10** arranged within the closed frame formed by the side elements **700**n. According to one example, the two outermost side elements **700**a, **700**d, in an assembled state of the housing **7**, may be simply glued or welded to each other. That is, the fastening element may comprise a glue bead or welding seam, for example. According to another example, the fastening element **710**, **712** may be any kind of snap-fit connector or latching mechanism. In the examples illustrated in FIGS. **4** to **9**, the fastening element **710**, **712** comprises a protrusion **710** arranged on the outside (in the assembled state) of the first outermost side element **700**a, and a kind of loop **712** attached to the fourth outermost side element **700**d. When the housing **7** has been folded to form a closed loop, the loop **712** may be folded to engage with the protrusion **710**. In this way, the two outermost side elements **700**a, **700**d may be securely attached to each other such that the housing **7** remains in its folded state. That is, the two outermost side elements **700**a, **700**d may be permanently attached to each other (e.g., by gluing or welding) such that they may not be easily separated again, or the two outermost side elements **700**a, **700**d can be separated again by simply opening the fastening element **710**, **712**. The latter is generally possible with many latching mechanisms which can be released again by applying a certain amount of force without damaging the fastening element **710**, **712** or the two outermost side elements **700**a, **700**d in any way.

[0037] In the embodiments described herein, the housing **7** in its folded state has a rectangular shape. This, however, is only an example. It is also possible that the housing **7** in its folded state has a square shape. That is, according to embodiments of the disclosure, n may be 4 and m may be 3 (n=4, m=3). It is, however, possible that the housing **7** comprises more (e.g., more than four) or less (e.g., three) side elements **700**n. That is, the housing **7** in its folded state generally may have any polygonal shape. As there are always two free ends (two outermost side elements), the number m of hinge portions **700**i is always one less than the number n of side elements **700**n.

[0038] The hinge portions **700**i may be implemented in many different ways. In the example illustrated in FIG. **4**, the hinge portions **700**i each comprise a continuous section of a flexible material. For example, the side elements **700**n may be formed by a comparably rigid (e.g., plastic) material in order to provide sufficient stability for the housing **7**. The side elements **700**n may further have a certain thickness d700, e.g., between 1 mm and 3 mm. The side elements **700**n, therefore, may not be easily bent without causing damage to the side elements **700**n. The hinge portions **700**i may be formed by applying a comparably flexible material to the edges of the respective side elements **700**n facing each other. The hinge portions **700**i, therefore, are flexible to a certain degree. In this way, they can be bent in order to form the closed frame. On the other hand, the flexible portions **700**i are still able to form a permanent connection between the respective ones of the side elements **700**n.

[0039] Another example of how the hinge portions **700**i can be implemented is schematically illustrated in FIG. **6**, wherein FIG. **6**A illustrates an outside of the housing **7**, FIG. **6**B illustrates a top view of the housing **7**, and FIG. **6**C illustrates an inside of the housing **7**. Each of the side elements **700**n has the first thickness d**700**$_{max}$, and each of the hinge portions **700**i comprises a first end having the first thickness d**700**$_{max}$ and connected to one of the side elements **700**n, and a second end having the first thickness d**700**$_{max}$ and connected to another one of the side elements **700**n. Each hinge portion **700**i further comprises a first section, a second section, and a third section arranged successively between the first end and the second end, wherein the second section has a second thickness d**700**$_{min}$ that is less than the first thickness d**700**$_{max}$. A thickness of the first section between the first end and the second section gradually decreases from the first thickness d**700**$_{max}$ to the second thickness d**700**$_{min}$, and a thickness of the third section between the second section and the second end gradually increases from the second thickness d**700**$_{min}$ to the first thickness d**700**$_{max}$. The housing **7** is configured to form a closed frame in an assembled shape by bending the second sections of the connecting portions **700**i about a respective folding axis until the respective first and third sections are in direct contact with each other. The hinge portions **700**i in this example may be formed of the same material as the side elements **700**n. That is, the hinge portions **700**i may also be made from a comparably rigid (e.g., plastic) material. As the thickness of the hinge portions **700**i is reduced to the second thickness d**700**$_{min}$ in their respective second sections, they may, however, still be bendable. That is, the second sections of the hinge portions **700**i may be thin enough in order to allow them to be bent sufficiently to form the closed loop. The second sections of the hinge portions **700**i remain intact also in the assembled state of the housing in order to couple the respective side elements **700**n to each other.

[0040] A folding process of the housing of FIG. **6** is schematically illustrated in FIGS. **7**A to **7**D, wherein FIG. **7**A illustrates the housing in its unfolded state, FIG. **7**D illustrates the housing in its assembled state, and FIGS. **7**B and **7**C illustrate intermediate steps between the unfolded and the assembled state. In FIGS. **7**A to **7**D, a substrate **10** is not specifically illustrated. The housing **7**, however, may be easily folded around a substrate **10**, resulting in an arrangement similar to what is illustrated in FIG. **3**. The shape of the housing **7** in its folded state (inner circumference) corresponds to the shape of the respective substrate (outer circumference). In this way, each side surface of the substrate **10** will be contacted by a respective one of the side elements **700**n of the housing **7**.

[0041] In the example illustrated in FIG. **4**, the side elements **700**n are essentially rectangular and two neighboring side elements **700**n directly contact each other. This results in a somewhat irregular shape of the housing **7** in the assembled state (see, e.g., FIG. **5**). The hinge portions **700**i as illustrated in FIG. **6** result in a more regular shape of the housing **7** in the assembled state. In the example of FIGS. **4** and **5**, the fourth outermost side element **700**d in the assembled state rests flat on the first outermost side element **700**a. This, however, is only an example. The housing **7** may further comprise a first connecting section **700**j, and a second connecting section **700**j. The first and second connecting section **700**j may each form an outermost end of the housing **7** in its unfolded state. This is schematically illustrated in FIGS. **6**A to **6**C and **7**A to **7**D. The first connecting section **700**j may comprise a first end having the first thickness d**700**$_{max}$ and being connected to the first outermost side element **700**a such that the first outermost side element **700**a is arranged between the first connecting section **700**j and the respective hinge portion **700**i connected to an opposite end of the first outermost side element **700**a. The thickness of the first connecting portion **700**j gradually decreases towards a second end of the first connecting portion **700**j facing away from the first outermost side element **700**a. The first connecting portion **700**j, for example, may have a triangular cross-section. The same applies for the second connecting portion **700**j, which may comprise a first end having the first thickness d**700**$_{max}$ and being connected to the fourth outermost side element **700**d such that the fourth outermost side element **700**d is arranged between the second connecting section **700**j and the respective hinge portion **700**i connected to an opposite end of the fourth outermost side element **700**d. The thickness of the second connecting portion **700**j gradually decreases towards a second end of the second connecting portion **700**j facing away from the fourth outermost side element **700**d. The second connecting portion **700**j, for example, may have a triangular cross-section. In the assembled state of the housing **7**, the first connecting portion **700**j and the second connecting portion **700**j may be in direct contact with each other, forming one of the corners of the housing **7** (see, e.g., FIG. 7D).

[0042] The housing **7**, i.e. the side elements **700**n, may have a height h**700** in a vertical direction y. When the housing **7** is folded around a substrate **10**, the substrate **10** may contact the side elements **700**n in an area towards a lower end of the side elements **700**n. A lower end of the side elements **700**n is an end opposite a top end to which a cover or lid may be mounted. For example, a distance d**10** (see FIG. **3**) between the substrate **10** and the lower end in the vertical direction y may be between **0** and **2** cm. In this way, the substrate **10** forms the bottom of the housing **7**. When the housing **7** is folded around the substrate **10** and the two outermost side elements **700**a, **700**d (and the first and second connecting portions **700**j, respectively) have been securely attached to each other by means of the fastening element **710**, **712**, a force is applied to the substrate **10** horizontally from all sides. In this way, the substrate **10** is clamped by the closed frame formed by the side elements **700**n. The substrate **10** may be in direct contact with the side element elements **700**n of the housing. However, there is a risk that the housing **7** is not entirely sealed.

[0043] Therefore, the housing **7** may further comprise a sealing element **720**. The sealing element **720**, in the assembled state of the housing **7**, extends around the inner volume defined by the first plurality n of side elements **700**n. A sealing element **720** is exemplarily illustrated in FIGS. **4** to **7** by means of a dashed line. As can be seen in FIGS. **4**B, **4**C, **6**B and **6**C, for example, the sealing element **720** may be discontinuous in the unassembled state of the housing **7**. In particular, the sealing element **720** may be interrupted in the areas of the hinge portions **700**i. That is, the sealing element **720** is attached to the side elements **700**n, but not to the hinge portions **700**i. The sealing element **720** may form a continuous loop along the inner circumference of the housing **7** in the assembled state, as is illustrated in FIGS. **5** and 7D, for example. The sealing element **720** is configured to be arranged between the substrate **10** and the side elements **700**n in the assembled state. According to one example, the sealing element **720** is formed by a comparably soft material (e.g., silicone material, or silicone rubber material). For example, the sealing element **720** may be a bead or bulge extending along the side elements **700**n in an area towards the lower end of the side elements **700**n, e.g., within 0 and 1 cm, or between 0 and 2 cm, from a lower end of the side elements **700**n. When the housing **7** is arranged to surround the substrate **10**, and the sealing element **720** is arranged between the substrate **10** and the side elements **700**n, the sealing element **720** is deformed to a certain degree due to the pressure that is applied by the housing **7**. That is, the substrate **10** presses into the sealing element **720** to a certain degree. In this way, the connection between the side elements **700**n and the substrate **10** is even tighter, resulting in a very reliable mechanical fixation of the substrate **10** with respect to the housing **7**. Further, the sealing element **720** may reliably prevent material from leaking out of the housing **7**, e.g., when a liquid or viscous material is filled into the housing **7** in order to form an encapsulant **5**. This is schematically illustrated in FIGS. **13**, **14** and **15**A, for example, where the substrate **10** presses into the sealing element **720** arranged between the substrate **10** and the side elements **700**n.

[0044] Referring to FIGS. **4**C and **6**C, for example, sections of the sealing element **720** may also extend in a vertical direction y of the side elements **700**n. In particular, one section of the sealing element **720** extending in the vertical direction y may be arranged on the first outermost side element **700**a, and/or one section of the sealing element **720** extending in the vertical direction y may be arranged on the fourth outermost side element **700**d. In this way, the housing **7** may be sealed also in a section where the first outermost side element **700**a and the fourth outermost side element **700**d are in direct contact with each other. That is, a vertical section of the sealing element **720** may extend along an end of the first outermost side element **700**a that faces away from the end connected to the respective hinge portion **700**i. Similarly, a vertical section of the sealing element **720** may extend along an end of the fourth outermost side element **700**d that faces away from the end connected to the respective hinge portion **700**i. In the assembled state of the housing **7**, the vertical sections of the sealing element **720** may be in direct contact with each other, thereby sealing the connection point between the two outermost side surfaces **700**a, **700**d.

[0045] In the examples illustrated in FIGS. **4** to **7**, the housing **7** comprises only sidewalls (side elements **700**n). A separate lid or cover may be arranged on the sidewalls to close the housing **7** after the housing **7** has been folded around the substrate **10**, for example. According to another example, the housing **7** may comprise a lid **730** that is coupled to (or integrally formed with) one of the plurality of side elements. In the example illustrated in FIGS. **8**A to **8**C, a lid **730** is coupled to a second one of the side elements **700**b. FIG. **8**A illustrates an outside of the housing **7**, FIG. **8**B illustrates a top view of the housing **7**, and FIG. **8**C illustrates an inside of the housing **7**. The lid **730** is foldable with respect to an additional folding axis. In the assembled state of the housing **7**, the closed frame formed by the first plurality n of side elements **700**n defines an inner volume. The lid **730**, which in an unassembled state of the housing **7** is arranged in the same plane as the plurality of side elements **700**n, may be bent about a folding axis with respect to the respective side element **700**b such that, in the assembled state of the housing **7**, it covers the inner volume from one side (i.e. from a top side opposite the lower side to which the substrate **10** is attached). In the assembled state, the lid **730** is arranged perpendicular to each of the plurality of side elements **700**n. The lid **730** may be attached to the respective side element **700**n in any suitable way which allows it to be bent about the respective folding axis. The lid **730**, for example, may be attached to the respective side element **700***n* by means of a suitable hinge element, which allows a bending of the lid **730** with respect to the side elements **700***n*. In the assembled state of the housing **7**, the lid **730** may be securely attached to the side elements **700***n* by means of suitable fastening elements **732**, **734**, similar to what has been described with respect to the two outermost side elements **700***a*, **700***d* and the respective fastening element **710**, **712** above. That is, the fastening elements **732**, **734** that attaches the lid **730** to the side elements **700***n* may comprise any kind of snap-fit connectors, for example.

[0046] Now referring to FIG. **9**, the lid **730** may be coupled to the respective one of the plurality of side elements **700***b* by means of a foldable element **736**. FIG. **9**A illustrates an outside of the housing **7**, FIG. **9**B illustrates a top view of the housing **7**, and FIG. **9**C illustrates an inside of the housing **7**. The foldable element **736** may be configured to be folded in an accordion-like way, which is schematically illustrated in FIG. **10** which illustrates a cross-sectional view of a housing **7** folded around a substrate **10**. The foldable element **736**, for example, may comprise two sections. A first section may be coupled to the respective side element **700***b*, and a second section may be coupled to the lid **730**. The first section may be bendable with respect to the side element **700***b*, the second section may be bendable with respect to the lid **730**, and the first and second section may be bendable with respect to each other. This allows a movement of the lid **730** that is linearly downward, instead of a rotary motion (as in the example of FIG. **8**). As has been described with respect to FIG. **1** above, a semiconductor module generally comprises terminal elements **4** extending vertically through the housing **7**. The lid **730** may comprise openings (not specifically illustrated in FIGS. **9**A and **9**C through which the terminal elements **4** may extend towards the outside of the housing **7**. Such terminal elements **4** are generally already arranged on the substrate **10** when the lid **730** is closed. A rotary movement of the lid **730** may potentially cause problems when inserting the terminal elements **4** through the respective openings in the lid **730**. Such problems may be avoided in the arrangement of FIG. **9**, due to the linearly downward movement of the lid **730**, as schematically illustrated by means of the arrows in FIG. **10**. FIG. **10**A schematically illustrates the housing **7** with the lid **730** almost completely open, and FIG. **10**B schematically illustrates the housing **7** with the lid **730** folded to a suitable horizontal position, from which it may then be moved linearly downward in a vertical direction y toward the side elements. FIG. **10**C schematically illustrates the housing **7** and the lid **730** in a state shortly before the lid **730** reaches its final mounting position on the side elements. The foldable element **736** in FIG. **10**C is almost completely folded. A section of the housing **7** is shown in an enlarged view in FIG. **10**C, in order to more clearly illustrate the foldable element **736** during the folding process. The foldable element **736** as illustrated in FIGS. **9** and **10**, however, is only an example. A linearly downward movement of the lid **730** towards the side elements **700***n* may also achieved in other ways.

[0047] Similar to what has been described with respect to the side elements **700***n* above, a sealing element **720** may also be attached to the lid **730**, for example. As can be seen in FIG. **9**C, for example, which illustrates an inside of the housing **7**, the sealing element **720** may form a continuous thread around the circumference of the lid **730** on the inside of the housing **7**. When the lid **730** is closed to cover the opening defined by the side elements **700***n* of the housing **7**, the sealing element **720** is arranged between the lid **730** and the side elements **700***n*, and seals any gaps between the lid **730** and the side elements **700***n*. Instead of on the lid **730**, the sealing element **720** may also be arranged on a top surface of the side elements **700***n*. A sealing element **720** arranged on a top surface of the side elements **700***n* would be visible in the top view of FIG. **9**B instead of in FIG. **9**C. In the latter case, however, the sealing element **720** would comprise several separate sections which, in the folded state of the housing **7** would form a continuous thread around the circumference of the housing **7**. When forming the sealing element **720** on the lid **730**, the thread formed by the sealing element **720** would have no disruptions at all which may provide even better sealing properties.

[0048] Further sealing elements may be provided for any gaps or holes that may be present in the assembled state of the housing **7**. As has been described above, a lid **730** for a housing **7** may comprise a plurality of holes through which the terminal elements **4** may protrude to the outside of the housing **7**. Further sealing elements may be provided for some or all of such openings in the lid **730**. According to one example, a layer of a comparably soft and flexible material may cover each of a plurality of openings in the lid **730**. When a terminal element **4** is driven through an opening, it may penetrate through this layer. In the assembled state, such a sealing element then tightly surrounds the respective terminal element **4**, thereby sealing the opening with the terminal element extending therethrough. Any openings through which no terminal element **4** is inserted remain covered by the respective layer of sealing material. In this way, the inside of the housing **7** may be entirely sealed to prevent any noxious gases (e.g., hydrogen or hydrogen sulphide) or contaminants from entering the inside of the housing **7**.

[0049] The housing **7** may be manufactured by means of an injection molding process, for example, by filling the material in a liquid or viscous form into a respective mold before hardening it. Conventional housings are often manufactured by means of injection molding processes. Conventional housings, however, are injection molded in their final form, as is schematically illustrated in the cross-sectional view of FIG. **11**A. As can be seen, each element **902***a*, **902***b* of the mold requires a significant amount of space, as the housings have comparably large dimensions in both horizontal directions x, z (dimensions of the lid), as well as in the vertical direction y (height of sidewalls). Now referring to FIG. **11**B, it can be seen that the dimensions of the housings **7** according to embodiments of the disclosure in their unassembled state significantly differ from the dimensions of conventional housings. In particular, the housings **7** in their unassembled state are comparably flat. This allows to manufacture several different housings **7** in parallel, by stacking respective molds onto each other. In the example illustrated in FIG. **11**B, three housings **7** are formed simultaneously with the same equipment. This, however, is only an example. It is generally possible to easily form less than three or even more than three housings **7** simultaneously with the same equipment, without requiring unusually large equipment. The molding form that is used to form the housings **7** may be a so-called multi-cavity injection molding form, comprising a plurality of different elements **902**_a_, **902**_b_, **902**_c_, **902**_d_ stacked onto each other.

[0050] Typically, a simple injection molding form consists of two elements **902**_a_, **902**_b_ which, during the molding process, are pressed to each other. A cavity defined by the two elements **902**_a_, **902**_b_ is a negative of the housing **7** to be formed. This cavity is filled with material which is subsequently hardened before disengaging the two elements **902**_a_, **902**_b_ from each other (see FIG. **11**A). In order to reduce production costs, a plurality of housings **7** may be formed in parallel by using multi-cavity forms. A size of the molding forms, however, is typically limited to a maximum of about 350 mm*350 mm; larger dimensions are possible but expensive. Therefore, the ability to create higher level multi-cavity forms may be limited. Using flat geometries, as is exemplarily illustrated in FIG. **11**B, offers the opportunity to create multi cavity forms that are arranged serially with respect to each other. With a very flat geometry of the injection molding element (the housing **7**), multi-cavity forms can be created by means of a serial multi-cavity injection molding form.

[0051] The injection molding process may be a so-called 2K LSR (liquid silicone rubber) process. This is a process where a first material is first filled into the mold and subsequently hardened, thereby forming a first injection molding component. The first injection molding component may be comparably rigid and hard after hardening, for example. For example, the side elements **700**_n_, hinge portions **700**_i_, fastening elements **710**, **712**, a lid **730**, foldable element **736**, as well as the first connecting section **700**_j_ and the second connecting section **700**_j_ may be formed during this first step. A second molding step follows during which one or more second injection molding components are formed. The second injection molding component may be the one or more sealing elements **720**, for example. The one or more second injection molding components may be formed by injecting a different material (e.g., a liquid silicone rubber) into the mold such that the one or more second injection molding components are formed directly on the first injection molding component (first material). The second injection molding components, after curing and hardening, adhere to the first injection molding component. This may be achieved by means of rotatable forms **902**_a_, **902**_b_, **902**_c_, **902**_d_, as schematically illustrated in FIG. **12**.

[0052] In the top part of FIG. **12**, the molding form is illustrated in a first position. The first injection molding component may be formed when the different parts **902**_a_, **902**_b_, **902**_c_, **902**_d_ are pressed to each other in this first position. In the example illustrated in FIG. **12**, four housings **7** are formed simultaneously with the same equipment. One or more of the mold parts **902**_a_, **902**_b_, **902**_c_, **902**_d_ may then be rotated. In FIG. **12**, the two outermost mold parts **902**_a_, **902**_d_ are rotated. When the mold parts **902**_a_, **902**_b_, **902**_c_, **902**_d_ are in a second position with respect to each other (FIG. **12** bottom part), the one or more second injection molding components may be formed, similar to what has been described above. A rotating axis about which the parts **902**_a_, **902**_d_ are rotated is illustrated in dot-dashed lines in FIG. **12**.

[0053] Now referring to FIG. **13**, it is even possible to form more sophisticated foldable housings. The housing **7** as illustrated in FIG. **13**, for example, further comprises a mounting element **740** which allows to securely mount the housing **7** on a heat sink **82**, for example. Such a mounting element **740** may comprise a protrusion with a through hole therethrough that allows a screw **72** to be inserted. FIG. **13** illustrates a very simple L-shape design of a mounting element **740** with respect to the respective side element **700**_n_. That is, the mounting element **740** may be static with respect to the side element **700**_n_. This, however, increases the dimensions of the housing **7** in its unfolded state to a certain degree.

[0054] FIG. **14** schematically illustrates another static element which can be used to mount the housing **7** onto a heat sink **82**, for example. In this example, a simple, comparably flat protrusion **750** is formed along the side element **700**_n_. The protrusion **750** may have a hole through which a first end of a spring element **752** (e.g., metal spring element) may be inserted. A second end of the spring element **752** may be attached to a heat sink **82** by means of a screw **72**, for example. The first end of the spring element **752** may be inserted into the protrusion **750** from a top side (facing towards the lid of the housing **7**). In this way the spring element **752** presses the housing **7** towards the heat sink **82** when the screw **72** is securely fastened.

[0055] Now referring to FIG. **15**, it is also possible to form foldable mounting elements. The folding mechanisms may generally resemble the folding mechanisms that have been described with respect to the hinge portions **700**_i_ and the lid **730** above. For example, a foldable mounting mechanism may comprise several different parts. A first part **760** may be attached to a side element **700**_n_. In the example of FIG. **15**, the first part **760** is attached to the side element **700**_n_ by means of a silicone rubber material **766**. This may be the same material that is used to form the sealing element **720**, for example, and may be formed during the same production step. A second part **762** may be foldably attached to the first part **760**. For example, the second part **762** may also be attached to the first part **760** by means of a silicone rubber material **768**. In the unassembled state, the housing **7** with the foldable mounting element is still flat, as can be seen in FIG. **15**B which exemplarily illustrates a section of an outside of the housing **7** (similar to FIGS. **4**A, **6**A, **8**A and **9**A, for example). A cross-sectional view of a section of a housing **7** in its assembled state is schematically illustrated in FIG. **15**A.

[0056] As can be seen, the side element **700**_n_, the first part **760**, and the second part **762**, in the assembled state, may form a triangle. The second part **762** may comprise a through hole through which a screw **72** may be inserted in order to securely attach the housing **7** to a heat sink **82**. In order to allow the screw **72** to be inserted into the through hole in the assembled state, the first part **760** may also comprise an opening, a plurality of arms or may be formed by two separate sections (as schematically illustrated in FIG. **15**B), in order to not cover the second part **762** and the through hole. The second part **762**, in the assembled state of the housing **7**, may be inserted (e.g., hooked) into a respective notch or groove in the side element **700**_n_. This makes the triangle formed by the different parts **760**, **762** of the mounting element and the side element **700**_n_ very stable. Further, the force that is applied to the second part **762** when the screw **72** is fastened, is transferred to the entire side element **700**_n_. The different mounting elements as have been described with respect to FIGS. **13** to **15** above, are only examples. Mounting elements may generally be implemented as static mounting elements or foldable mounting elements in any suitable way.

[0057] As used herein, the terms "having", "containing", "including", "comprising" and the like are open ended terms that indicate the presence of stated elements or features, but do not preclude additional elements or features. The articles "a", "an" and "the" are intended to include the plural as well as the singular, unless the context clearly indicates otherwise.

[0058] The expression "and/or" should be interpreted to include all possible conjunctive and disjunctive combinations, unless expressly noted otherwise. For example, the expression "A and/or B" should be interpreted to mean only A, only B, or both A and B. The expression "at least one of" should be interpreted in the same manner as "and/or", unless expressly noted otherwise. For example, the expression "at least one of A and B" should be interpreted to mean only A, only B, or both A and B.

[0059] It is to be understood that the features of the various embodiments described herein can be combined with each other, unless specifically noted otherwise.

[0060] Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations can be substituted for the specific embodiments shown and described without departing from the scope of the present invention. This application is intended to cover any adaptations or variations of the specific embodiments discussed herein. Therefore, it is intended that this invention be limited only by the claims and the equivalents thereof.

What is claimed is:

1. A housing for a semiconductor module comprises:

a first plurality (n) of side elements; and

a second plurality (m) of hinge portions, wherein:

$n=m+1$;

in an unassembled state of the housing, the side elements are arranged successively in one row and each of the hinge portions is arranged to connect two directly neighboring side elements;

each of the side elements is foldable with respect to at least one other side element about a respective folding axis; and

the side elements are configured to form a closed frame in an assembled state of the housing.

2. The housing of claim **1**, wherein each of the hinge portions is formed of a silicone rubber material.

3. The housing of claim **1**, wherein

each of the side elements has a first thickness;

each of the hinge portions comprises a first end having the first thickness and connected to one of the side elements, a second end having the first thickness and connected to another one of the side elements, a first section, a second section, and a third section arranged successively between the first end and the second end, wherein the second section has a second thickness that is less than the first thickness, wherein a thickness of the first section between the first end and the second section gradually decreases from the first thickness to the second thickness, and wherein a thickness of the third section between the second section and the second end gradually increases from the second thickness to the first thickness; and

the housing is configured to form a closed frame in an assembled shape by bending the second sections of the connecting portions about a respective folding axis until the respective first and third sections are in direct contact with each other.

4. The housing of claim **1**, further comprising a lid coupled to one of the plurality of side elements, and foldable with respect to an additional folding axis.

5. The housing of claim **4**, wherein, in the assembled state of the housing, the closed frame formed by the first plurality (n) of side elements defines an inner volume, and the lid covers the inner volume from one side.

6. The housing of claim **4**, wherein the lid is coupled to the respective one of the plurality of side elements by means of a foldable element, wherein the foldable element is configured to be folded in an accordion-like way.

7. The housing of claim **4**, wherein, in the assembled state of the housing, the lid is attached to the closed frame formed by the side elements by means of one or more fastening elements.

8. The housing of claim **1**, further comprising a sealing element, wherein the sealing element, in the assembled state of the housing, extends around the inner volume defined by the side elements.

9. The housing of claim **8**, wherein the sealing element further comprises a portion forming a continuous thread around the circumference of the lid such that, in the assembled state of the housing, it is arranged between the lid and the side elements, thereby sealing the inner volume of the housing.

10. The housing of claim **1**, further comprising a fastening element, wherein the fastening element, in the assembled state of the housing, is configured to connect two outermost side elements in the row of side elements with each other such that the housing remains in its assembled shape.

11. The housing of claim **10**, wherein the fastening element is a snap-fit connector or latching mechanism.

12. The housing of claim **1**, further comprising at least one foldable mounting element attached to a side element, wherein the foldable mounting element comprises a through hole for mounting the housing on a heat sink by means of a screw.

13. The housing of claim **1**, wherein n=4 and m=3.

14. A semiconductor module comprising:

a housing (**7**), comprising:

a first plurality (n) of side elements; and

a second plurality (m) of hinge portions, wherein:

$n=m+1$;

in an unassembled state of the housing, the side elements are arranged successively in one row and each of the hinge portions is arranged to connect two directly neighboring side elements;

each of the side elements is foldable with respect to at least one other side element about a respective folding axis; and

the side elements are configured to form a closed frame in an assembled state of the housing; and

a substrate, wherein the housing forms a closed frame around the substrate and exerts pressure on each of a plurality of side surfaces of the substrate along the circumference of the substrate.

15. A method for assembling a semiconductor module, the method comprising:

providing a housing, comprising:
a first plurality (n) of side elements; and
a second plurality (m) of hinge portions, wherein:

$$n=m+1;$$

in an unassembled state of the housing, the side elements are arranged successively in one row and each of the hinge portions is arranged to connect two directly neighboring side elements;
each of the side elements is foldable with respect to at least one other side element about a respective folding axis; and
the side elements are configured to form a closed frame in an assembled state of the housing; and
folding the housing around a substrate by bending the plurality of side elements about respective bending axes until the side elements form a closed frame around the substrate and exert pressure on each of a plurality of side surfaces of the substrate along the circumference of the substrate.

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